

Foundation (Jalapeno)

Q1. What is the formula for solving $ax^2 + bx + c = 0$?

- (A) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 (B) $x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$
 (C) $x = \frac{-b \pm \sqrt{b^2 - 2ac}}{2a}$
 (D) $x = \frac{-b \pm \sqrt{b^2 + 2ac}}{2a}$
 (E) $x = \frac{-b \pm \sqrt{b^2 - ac}}{2a}$

Q2. What does the discriminant $b^2 - 4ac$ represent in a quadratic equation?

- (A) Number of solutions
 (B) Type of solutions
 (C) Both A and B
 (D) None of the above
 (E) Only real solutions

Q3. If the discriminant is positive, what type of solutions will the equation have?

- (A) One real solution
 (B) Two real solutions
 (C) Two complex solutions
 (D) One complex solution
 (E) No real solutions

Q4. What is the value of the discriminant for the equation $x^2 + 4x + 4 = 0$?

- (A) 0
 (B) 4
 (C) 8
 (D) 12
 (E) 16

Q5. If the discriminant is zero, how many solutions will the equation have?

- (A) One
 (B) Two
 (C) Three
 (D) Four
 (E) None

Q6. For the equation $2x^2 - 5x - 3 = 0$, what is the value of $b^2 - 4ac$?

- (A) $25 + 24 = 49$
 (B) $25 - 24 = 1$
 (C) $25 + 24 = 49$
 (D) $-25 - 24 = -49$
 (E) $25 - 24 = 1$

Q7. What is the number of solutions for the equation $x^2 + 2x + 2 = 0$?

- (A) One real solution
 (B) Two real solutions
 (C) Two complex solutions
 (D) No real solutions
 (E) Infinite solutions

Q8. If the discriminant is negative, what type of solutions will the equation have?

- (A) One real solution
 (B) Two real solutions
 (C) Two complex solutions
 (D) One complex solution
 (E) No solutions

Open Ended Questions (FRQ)

Q9. Solve the equation $x^2 - 6x + 8 = 0$ using the quadratic formula. Show all work and explain the number and type of solutions.

Q10. Find the discriminant of the equation $3x^2 + 2x - 5 = 0$. Then, determine the number and type of solutions based on the value of the discriminant.

Chef's Tips

- Provide a formula sheet with the quadratic formula.
- Emphasize the importance of checking the discriminant before solving.
- Remind students to show all work and explain their reasoning.